

Erdős Problem #256

EP-256

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $n \geq 1$ and $f(n)$ be maximal such that for any integers $1 \leq a_1 \leq \dots \leq a_n$ we have $\max_{|z|=1} |\prod_i (1 - z^{a_i})| \geq f(n)$. Estimate $f(n)$ - in particular, is it true that there exists some constant $c > 0$ such that $\log f(n) \gg n^c$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>